

$$P < 0.05$$

$$Q$$

$$p \gg n$$

p

10×10

10×10

$Xn \times p$

$$X = U \Sigma V^T$$

U

$\Sigma \sigma$

V

$$X^T X V \Sigma^2$$

$$\lambda_i = \frac{\sigma_i^2}{n-1}$$

$$d_{ij}||z_i-z_j||$$

$$=\sqrt{\sum_{i,j}(d_{ij}-||z_i-z_j||)^2}$$

$$n=500, p=10,000 p \gg n$$

$$\begin{array}{c} X \\ D \end{array}$$

$$\begin{array}{l} p \gg n \\ X = U \Sigma V^T \end{array}$$

W	
L	$D - W$

Wd
 L
 D
 L

$L = D - W$

LL
$$N$$
$$N \rightarrow N - 1 \rightarrow \cdots \rightarrow 1$$
 K

$$W$$
$$L$$
$$N \times NN > 10,000$$
 K

p

$$P < 0.05$$

α	
	$1/N$

$$P < 0.05$$

$$0.0010.1$$

$$m = 20,000$$

$$\alpha = 0.05$$

$$20,000 \times 0.05 = 1,000$$

$$N$$

$$\alpha_{new} = \frac{\alpha}{m}$$

$$\alpha = 0.05$$

$$P_{threshold} = \frac{0.05}{20,000} = 0.0000025$$

$$P_{(1)} \leq P_{(2)} \leq \cdots \leq P_{(m)}$$

$$k$$

$$P_{(k)} \leq \frac{k}{m} \times \alpha$$

$$k$$

$$k$$

$$1k$$

$$m = 5\alpha = 0.05[0.001, 0.009, 0.04, 0.045, 0.6]$$

k	$P_{(k)}$	$\frac{k}{5} \times 0.05$	$P \leq$
		0.01	✓
		0.02	✓
		0.03	×
			×

$$P = [0.001, 0.015, 0.025, 0.05, 0.6]\alpha = 0.05$$

$$k = 10.001 \leq 0.01$$

$$k = 20.015 \leq 0.02$$

$$k = 30.025 \leq 0.03 \rightarrow$$

$$k = 40.05 > 0.04$$

$$0.01k = 1$$

$$0.05/10,000 = 0.000005$$



$$y = x$$

$$\alpha/m$$

$$P_{(k)} \leq \frac{k}{m} \alpha$$

A	
	A
d	

$$C_D(v) = (v) = v$$

$$(s,t)v$$

$$C_B(v) = \sum_{s \neq v \neq t} \frac{\sigma_{st}(v)}{\sigma_{st}}$$

$$\sigma_{st}s \rightarrow t\sigma_{st}(v)v$$

x_ix_j

$$x_i = \frac{1}{\lambda} \sum_{j \in (i)} x_j \iff Ax = \lambda x$$

d

$C(T_i)1/100$

$$1/N$$

Q

Q	
	λ_2

$\updownarrow$

Q

$$Q = \frac{1}{2m} \sum_{i,j} \left(\underbrace{A_{ij}} - \underbrace{\frac{k_i k_j}{2m}} \right) \delta(c_i, c_j)$$

$$A_{ij}i,j$$

$$\frac{k_i k_j}{2m}ijk$$

$$\delta(c_i, c_j)ij$$

$$m=10$$

$$A_{AB}$$

$$E^{\frac{4\times 5}{2\times 10}}=\frac{20}{20}=1$$

$$1-1=0$$

$$E^{\frac{1\times 1}{20}}=0.05$$

$$1-0.05=0.95$$

$$L=D-A\lambda_2v_2$$

$$v_2=[0.1,0.2,-0.5,\ldots]$$

$$Lv_2$$

$$v_2$$

$$0$$

$$\approx \lambda_2$$

$$\lambda_2$$

$$Q$$

$$Q\approx 0.3\sim 0.7Q$$

$$v_2=L\lambda_1v_1[1,1,\ldots,1]\lambda_2$$

$$Q$$

$$v_2$$

Q

$G(n,p)$	
	k

p
 $G(n,p)np$
 $\lambda = np$
 k

$p_{in}p_{out}$

$$kP(k)k^{-\gamma}L$$

$$\rightarrow$$

$$\rightarrow$$

$$\rightarrow$$

$$-\gamma$$

$$P$$

$$P$$

$$tx,y$$

	$tt-k$

$$Y_t = \underbrace{T_t} + \underbrace{S_t} + \underbrace{R_t}$$

$$Y_tT_tS_tR_t$$

$$IY_t - Y_{t-1}$$

$$Y_t - Y_{t-7}$$

$$p$$

$$Y_t = \phi Y_{t-1} + \epsilon_t$$

$$d$$

$$q$$

$$Y_t = \epsilon_t + \theta \epsilon_{t-1}$$

$$\epsilon_{t-1}$$

$$pq$$

$$x_t$$

$$z_t$$

$$z_t$$

$$t-1$$

$$tx,y$$

$$Y = T + S + R$$

k	

$$y = wx + bwf(x)$$

$$f(x) \sim GP(m(x), k(x, x'))$$

$$m$$

$$kxx'f(x)f(x')$$

$$x_0Z(x_i)$$

$$\hat{Z}(x_0) = \sum_{i=1}^n \lambda_i Z(x_i)$$

$$\lambda_i \lambda$$

h

$n = 10,000$

± 100

W	
Y	

$P(Y|X)$

XY

$P(Y|(X))$

XY

$$i$$

$$Y_i(1)W = 1$$

$$Y_i(0)W = 0$$

$$\tau_i = Y_i(1) - Y_i(0)$$

$$Y_i^{obs} = W_iY_i(1) + (1 - W_i)Y_i(0)$$

$$Y_i(1)Y_i(0)$$

$$= \mathbb{E}[Y(1) - Y(0)] = \mathbb{E}[Y(1)] - \mathbb{E}[Y(0)]$$

$$\underbrace{\mathbb{E}[Y|W = 1] - \mathbb{E}[Y|W = 0]}_{\underbrace{\hspace{1cm}}} = \underbrace{\hspace{1cm}} + \underbrace{(\mathbb{E}[Y(0)|W = 1] - \mathbb{E}[Y(0)|W = 0])}_{\underbrace{\hspace{1cm}}}$$

$$W = 1W = 0Y(0)$$

$$W = 1Y(0)W = 0$$

$$10-3=+7$$

$$Y(1)$$

$$Y(0)$$

$$10-9=+1$$

$\neq$

	$Z \rightarrow X, Z \rightarrow Y$
	XY

$$e(x) = P(T = 1|X = x)$$

$$ZXYZ \rightarrow X \rightarrow Y$$

$$XY$$

$$ZX$$

$$ZX\hat{X}\rightarrow X$$

$$\hat{X}Y$$

$$XY$$

$$Z$$

$$Z \rightarrow X$$

$$Y$$

$$Z$$

Z	

$1024 \times 768500 \times 500224 \times 224$

$0 \sim 2550 \sim 1$

$\pm 15^\circ$

$x + F(x)$

$XZ\hat{X}Loss$

$$XZ$$

$$Z$$

$$Z\hat{X}$$

$$||X-\hat{X}||^2$$

$$\rightarrow Z$$

$$\rightarrow$$

$$\rightarrow$$

$$224 \times 224$$

Z

$$3 \times 3$$

$$\rightarrow$$

$$\rightarrow$$

$$H^{(l+1)} = \sigma(\underbrace{\tilde{D}^{-\frac{1}{2}} \tilde{A} \tilde{D}^{-\frac{1}{2}}}_{\text{}} H^{(l)} W^{(l)})$$

$$H^{l+1}AH^lW$$

α
